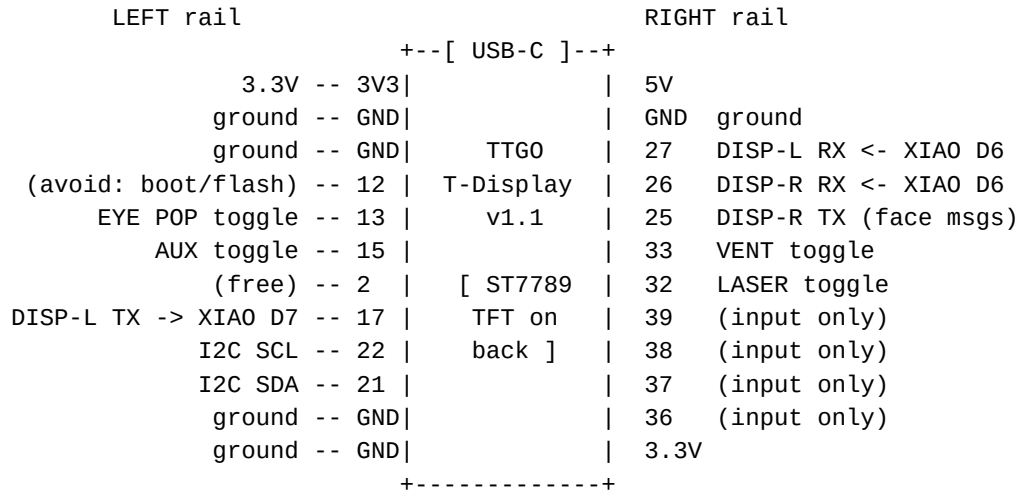


j4_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)



on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO00,
 GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
 I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, ADS_03 @ 0x4A, ADS_04 @ 0x4B,
 PCF8574 keypad_left @ 0x21, keypad_right @ 0x20
 toggles: INPUT_PULLUP, switch closes to GND (ON = LOW)

GPIO	Function
21	I2C SDA (ADS1115 x4, PCF8574 keypads x2)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, face-preset messages)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
32	LASER toggle (INPUT_PULLUP, switch closes to GND)
33	VENT toggle (INPUT_PULLUP, switch closes to GND)
13	EYE POP toggle (INPUT_PULLUP, switch closes to GND)
15	AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in; data / MAC / connection status)

j4_controller -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

ADS_01 @ 0x48 (ADDR -> GND)

both joysticks

+-----+

VDD	3.3V	
GND	GND	
SCL	TTGO GPIO 22 (I2C SCL)	
SDA	TTGO GPIO 21 (I2C SDA)	
ADDR	GND = address 0x48	
ALRT	not connected	
A0	neck joystick X wiper	-> neck L/R differential mix
A1	neck joystick Y (jaw) wiper	-> neck L/R base height
A2	eyes joystick X wiper	-> eyes pan servo (PCA9685 ch 3)
A3	eyes joystick Y wiper	-> eyes tilt servo (PCA9685 ch 4)

+-----+

ADS_02 @ 0x49 (ADDR -> VDD)

four face pots, left bank

+-----+

VDD	3.3V	
GND	GND	
SCL	TTGO GPIO 22 (I2C SCL)	
SDA	TTGO GPIO 21 (I2C SDA)	
ADDR	VDD = address 0x49	
ALRT	not connected	
A0	Eyebrow L pot wiper	-> PCA9685 ch 6
A1	Eyebrow R pot wiper	-> PCA9685 ch 7
A2	Basket Eyebrow L pot wiper	-> PCA9685 ch 8
A3	Basket Eyebrow R pot wiper	-> PCA9685 ch 9

+-----+

ADS_03 @ 0x4A (ADDR -> SDA)

three face pots, right bank

+-----+

VDD	3.3V	
GND	GND	
SCL	TTGO GPIO 22 (I2C SCL)	
SDA	TTGO GPIO 21 (I2C SDA)	
ADDR	SDA = address 0x4A	
ALRT	not connected	
A0	Nose pot wiper (up/down)	-> PCA9685 ch 10
A1	FAULTY channel, nothing connected, left unread in firmware	
A2	Bottom Eyelid L pot wiper	-> PCA9685 ch 12
A3	Bottom Eyelid R pot wiper	-> PCA9685 ch 13

+-----+

ADS_04 @ 0x4B (ADDR -> SCL)

neck-pivot + faders + Nose Basket

+-----+

VDD	3.3V	
GND	GND	
SCL	TTGO GPIO 22 (I2C SCL)	
SDA	TTGO GPIO 21 (I2C SDA)	
ADDR	SCL = address 0x4B	
ALRT	not connected	
A0	neck-pivot pot wiper	-> nP on the stepper link (0-3200)
A1	fader_left wiper	-> neck pivot (doubles ADS_04 A0, 0-3200)
A2	fader_right wiper	-> iris (doubles j4_display_right IRIS)
A3	Nose Basket pot wiper	-> PCA9685 ch 11 (single source)

+-----+